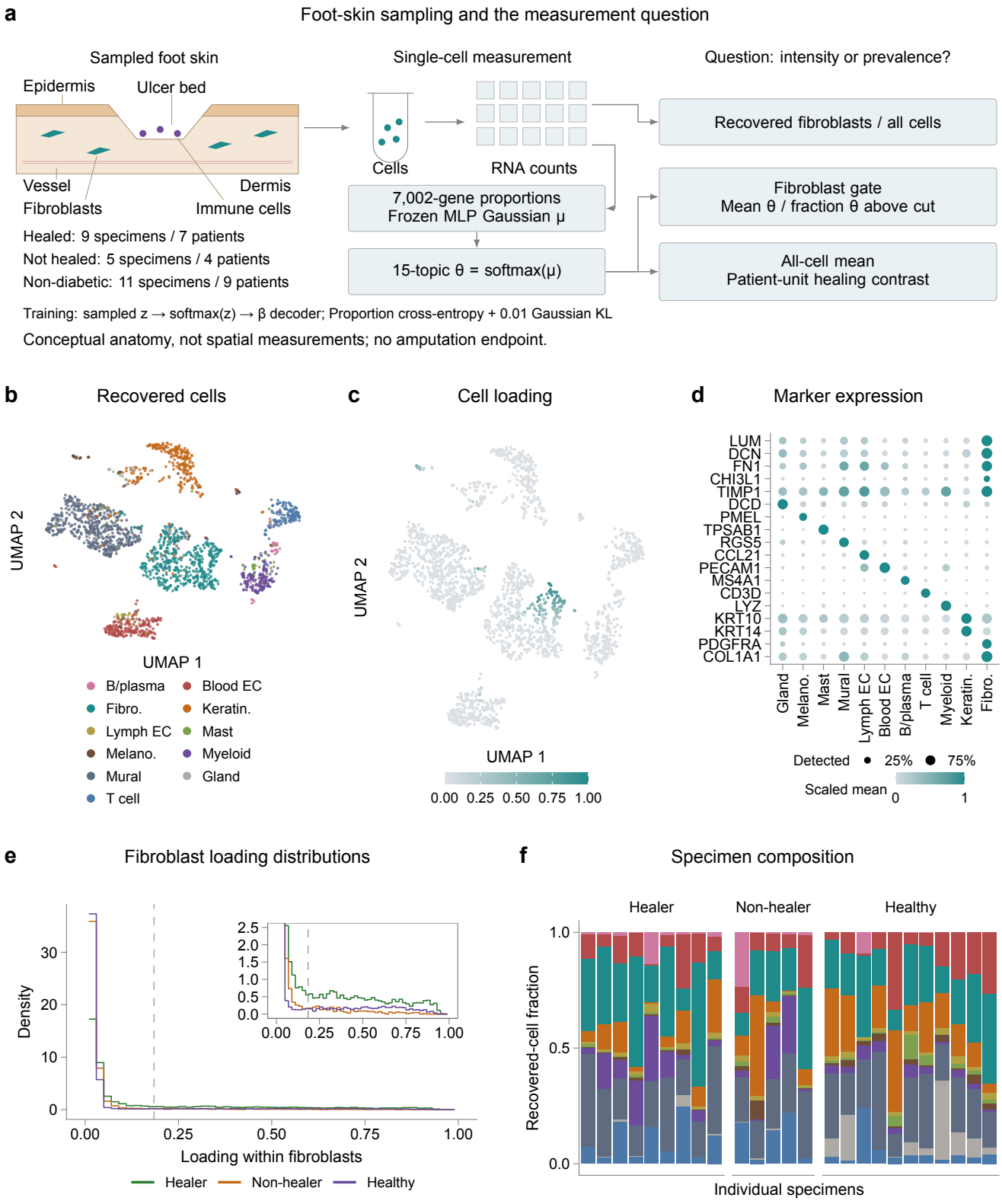

a
Foot-skin sampling and the measurement question
Sampled foot skin
Epidermis
Ulcer bed
Vessel
Dermis
Fibroblasts
Immune cells
Healed: 9 specimens / 7 patients
Not healed: 5 specimens / 4 patients
Non-diabetic: 11 specimens / 9 patients
Single-cell measurement
Cells
RNA counts
7,002-gene proportions
Frozen MLP Gaussian μ
15-topic θ = softmax(μ)
Question: intensity or prevalence?
Recovered fibroblasts / all cells
Fibroblast gate
Mean θ / fraction θ above cut
All-cell mean
Patient-unit healing contrast
Training: sampled z → softmax(z) → β decoder; Proportion cross-entropy + 0.01 Gaussian KL
Conceptual anatomy, not spatial measurements; no amputation endpoint.
b
Recovered cells
UMAP 2
UMAP 1
B/plasma
Blood EC
Fibro.
Keratin.
Lymph EC
Mast
Melano.
Myeloid
Mural
Gland
T cell
c
Cell loading
UMAP 2
UMAP 1
0.00 0.25 0.50 0.75 1.00
d
Marker expression
LUM
DCN
FN1
CHI3L1
TIMP1
DCD
PMEL
TPSAB1
RGS5
CCL21
PECAM1
MS4A1
CD3D
LYZ
KRT10
KRT14
PDGFRA
COL1A1
Gland
Melano.
Mast
Mural
Lymph EC
Blood EC
B/plasma
T cell
Myeloid
Keratin.
Fibro.
Detected 25% 75%
Scaled mean 0 1
e
Fibroblast loading distributions
Density
Loading within fibroblasts
Healer
Non-healer
Healthy
f
Specimen composition
Healer
Non-healer
Healthy
Recovered-cell fraction
Individual specimens